CO_2

Assessment Tool - 3

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Course Name : Computer Vision

ey Level Transformation :

Method : Log transformation

Log transformation is used to brighten dark images and reveal details in shadow regions.

Formula :

$$S = c \log (1+r).$$

Where,

$r \rightarrow$ Input intensity

$S \rightarrow$ output intensity

$C \rightarrow$ constant

Processing Steps :

1) Read the dark museum photograph.
2) Apply log transform to each pixel.
3) Normalize the output values to the range 0-255.
4) Display the enhanced image.

Result : Dark region become brighter, and hidden details of the exhibits become visible.

3) Histogram Equalization :

It improves the contrast of a fingerprint image by redistributing its intensity values.

Formula :

$$S_k = (L-1) \sum_{j=0}^{K} p(r_j).$$

Where.
L → Intensity levels
P(rj) → probability of intensity rj.

Processing Steps:
1) Read the fingerprint image.
2) Calculate the histogram of pixel intensities
3) Calculate the probability and cumulative distribution function.
4) Map the original intensities to new intensity values using the CDF.
5) Display the enhanced image.

Result: The contrast between fingerprint ridges and valleys increases, making the fingerprint clearer.

3) Salt - and - Pepper Noise Removal:
Method: Median filter.
A median filter removes salt and pepper noise while preserving object edges.
Processing Steps:
1) Read the noisy image.
2) Select a 3×3 neighbourhood around each pixel.
3) Arrange the nine pixel value in ascending order.
4) Replace the centre pixel with the median value.
5) Repeat the process for all pixels.

Why is it better than an averaging filter?

An averaging filter may blur edges because it calculates the mean of neighbouring pixels. A median filter removes extreme black and white noise values while preserving edges more effectively.

Result: Noise is reduced and object boundaries remain relatively sharp.

Image Sharpening:

Method: Laplacian Sharpening

Laplacian sharpening enhances edges and fine details in a blurred PCB image.

Laplace Mask:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Formula:

$$g(x,y) = f(x,y) - \nabla^2 f(x,y).$$

Processing Step:

1) Read the blurred PCB image.
2) Apply the laplacian mask to detect rapid intensity changes.
3) Obtain the laplacian response.
4) Subtract the response from the original image.
5) Display the sharpened image.

Result: Copper tracks and their edges become
and easier to inspect.

5) Frequency - Domain Low - Pass Filtering:

Method: Gaussian Low - Pass Filter

A Gaussian low pass filter reduces high - frequency noise and smooths the drone image.

Formula:

$$H(U,V) = e^{\frac{-D^2(U,V)}{(2D_0^2)}}$$

Where

$D(U,V)$ → distance from the frequency origin

D_0 → cutoff frequency.

Processing Steps:

1) Read the noisy drone image.
2) Apply the Fourier Transform.
3) Multiply the transformed image by the Gaussian low-pass filter.
4) Apply the Inverse Fourier Transform.
5) Display the filtered image.

Effect of cutoff Frequency:

Low cutoff: Remove more noise but causes more blurring

High cutoff: Preserves more detail but allow more noise.

easier to identify. ... blurred and ... with ...

Frequency Domain High Pass Filtering:

Method: High pass filter.

A high pass filter enhance fine details and edges, helping to reveal thin blood vessels in retinal images.

Formula:

$$H_{HP}(u,v) = 1 - H_{LP}(u,v).$$

Processing Step:

1) Read the retinal fundus image
2) Apply the fourier transform
3) Design a high pass filter.
4) Multiply the transformed image by the filter.
5) Apply the Inverse Fourier Transform and adjust the contrast.

Result: Thin blood vessels and fine edges become more visibile against the background.

Note: High pass filtering can amplify noise, so a mild filter should be used.

Point and line Detection :

Point detection identifies isolated air bubbles, while line detection identifies thin cracks on a glass bottle.

Point detection mask:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}.$$

Processing Steps:

1) Convert the bottle image into grayscale.
2) Apply the point detection mask to identify isolated points.
3) Apply horizontal, vertical and diagonal line mask to detect cracks.
4) Calculate the mask response for each pixel.
5) Compare the absolute response with a threshold T.

Threshold Selection : Select the threshold accordingly to the noise level and feature strength . A higher threshold reduces false detection but many miss faint cracks.

Result: Small air bubbles and hairline cracks are identified on the bottle surface.

Method: Adaptive Thresholding

Adaptive thresholding separate dark handwriting from a white page when the image has uneven illumination

Formula:

$$g(x,y) = \begin{cases} 0, & f(x,y) \leq T(x,y) \\ 255, & f(x,y) > T(x,y) \end{cases}$$

Where,

$f(x,y)$ → original intensity

$T(x,y)$ → local threshold.

Processing steps:

1) Read the handwritten answer-sheet image.
2) Convert it into grayscale.
3) Divide the image into small neighbourhoods.
4) Calculate a local threshold for each neighbourhood.
5) Convert pixels into black and white using the threshold.

Why adaptive thresholding?

Global thresholding uses one threshold for the whole image. Adaptive thresholding uses different local thresholds and handles uneven lighting better.

Result: Handwritting is separated clearly from the background.

9) Region Growing:

Region growing extracts a lake from a satellite image by grouping neighbouring pixels with similar intensity values.

Processing Steps:

1) Read the satellite image.
2) Select a seed point inside the lake.
3) Note the intensity of the seed point.
4) Examine neighbouring pixels and add those satisfying the similarity criterion.
5) Repeat until no more suitable pixels can be add.
6) Display the segmented lake region.

Seed point and similarity criterion:

Choose a seed point inside the lake, away from its boundary. Select an intensity threshold based on the difference between lake pixels and surroundings land pixels.

Result: The lake is separated from the surrounding land as a distinct region.

... connects broken edge segments, while the Hough Transform ... boundary detection identifies the complete outline of mechanical gears.

Processing Steps:

1) Read the gear image and convert it into grayscale.
2) Reduce noise using a suitable smoothing filter.
3) Apply an edge detector such as the Canny detector.
4) Use edge linking based on pixel proximity and gradient direction to connect related edges segment.
5) Apply the Hough transform to detect suitable lines or circular structures.
6) Use boundary tracing to obtain the gear outline.

Result: Broken edge segment are connected and the gear boundary becomes more complete for inspection

Note:

The hough transform is useful for straight lines and circular structures. Local edge linking and contour tracing are also important for irregular gear boundaries.